Expectations Formation with Fat-Tailed Processes: Evidence and Theory

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Motivation

- Large recent literature documenting **predictability** in expectation errors
  - Underreaction: lab + field, often short-term or consensus forecasts
  - Overreaction: lab + field, often longer-term individual forecasts

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- **This paper**: study expectations formation in the presence of "**fat**" **tails**
  - **Takeaway**: helps match data + parsimonious model of under & overreaction

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⇒ Allowing for **fat tails** is helpful for understanding belief formation!

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Related Literature

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   - Underreaction: lab Benjamin 19, ST earnings Bouchaud et al. 19, revenues Ma et al. 2024, ST rates Wang 21, macro (consensus) Coibion-Gorodnichenko 15
   - Overreaction: lab Afrouzi et al. 23, LT earnings growth Bordalo et al. 19, LT rates Giglio-Kelly 18, d'Arienzo 20, macro (individual) Bordalo et al. 20
   - **Contributions**:
     1. Field: evidence of both within **same** forecasting variable + horizon
     2. Lab: non-linearity in overreaction depends on the **Pareto tail** of DGP

Related Literature

1. Empirical evidence on under and overreaction in expectations
2. Models of under **or** overreaction in **individual** expectations
   - Underreaction: sticky expectations Bouchaud et al. 19, behavioral inattention Gabaix 19
   - Overreaction: diagnostic expectations Bordalo et al. 19, availability Afrouzi et al. 23

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1. Empirical evidence on under and overreaction in expectations
2. Models of under **or** overreaction in **individual** expectations
3. Models of under **and** overreaction in **individual** expectations
   - Experience effects/constant-gain learning Malmendier-Nagel 16, Nagel-Xu 19
   - Selective recall with similarity and interference Bordalo et al. 22, 23
   - Shrinkage towards average persistence or precision Wang 21, Augenblick et al. 24
   - Overreaction to category-specific features Kwon-Tang 25
   - Within vs. across-category comparisons Graeber et al. 24
   - **Contribution**: model with under + overreaction **within category/DGP**

RELATED LITERATURE

1. Empirical evidence on under and overreaction in expectations
2. Models of under **or** overreaction in **individual** expectations
3. Models of under **and** overreaction in **individual** expectations
4. Models of expectations with **unknown**/misspecified DGPs
   - Natural expectations Fuster et al. 10, 11
   - Learning Kozlowski et al. 20, Singleton 21, Farmer et al. 24, Dew-Becker et al. 24
   - No restrictions on the DGP de Silva-Thesmar 24
   - **Our focus**: misspecified model of distribution in the tails (could come from learning)

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4. Models of expectations with **unknown**/misspecified DGPs
5. Statistical models with **non-Gaussian** dynamics
   - Pareto tails, especially in firm growth Gabaix 09, Stanley et al. 96, Moran et al. 24
   - Skewness + kurtosis in income Guvenen et al. 14, 21, Braxton et al. 25
   - **Contribution**: connect with models of belief formation in a tractable way

Outline

1. Three Key Facts
   - Fact 1: Non-Linear Error-Revision Relationship
   - Fact 2: Fat Tails in the Distribution of Realizations
   - Fact 3: Expected Realizations are Non-Linear
2. Model of Expectations Formation
3. Additional Evidence
   - Forecasting Experiment
   - Macro Forecasts
   - Return Momentum
4. Conclusion

- **Sample**: 2000–2023 annual panel from LSEG IBES Detail files
- **Forecasting variables**: all available IBES firm-level outcomes

$$g_{it+h} = \begin{cases} \log X_{it+h} - \log X_{it+h-1} & \text{19 accounting items/per-share variables} \\ X_{it+h} & \text{3 accounting ratios} \end{cases}$$

Data and Variables

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- **Forecasts**:

$$F_t^j g_{it+h} = \begin{cases} \log F_t^j X_{it+h} - \log F_t^j X_{it+h-1} & \text{19 accounting items/per-share variables} \\ F_t^j X_{it+h} & \text{3 accounting ratios} \end{cases}$$

  - Main analysis uses analyst-level forecasts, but similar results with consensus
  - Ignores a Jensen's term, but results similar with % growth

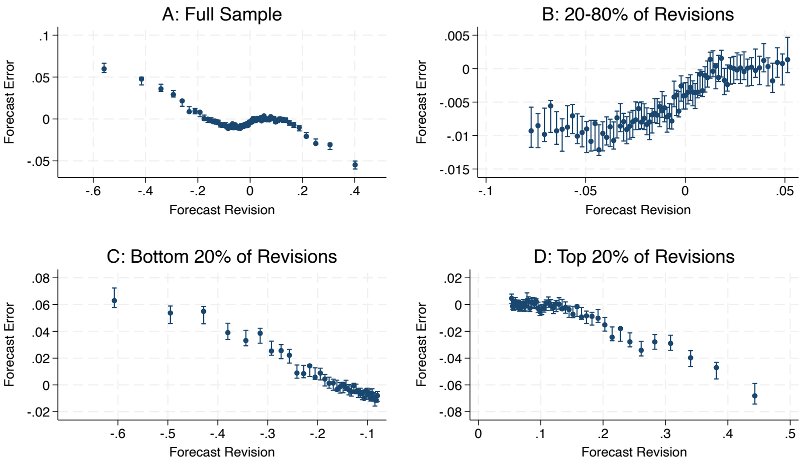
$$\underbrace{g_{it+1} - F_t^j g_{it+1}}_{\text{forecast error}} = \alpha + \beta \underbrace{\left(F_t^j g_{it+1} - F_{t-1}^j g_{it+1} \right)}_{\text{forecast revision}} + \epsilon_{it+1}^j$$

$$\underbrace{g_{it+1} - F^j_t g_{it+1}}_{\text{forecast error}} = \alpha + \beta \underbrace{\left(F^j_t g_{it+1} - F^j_{t-1} g_{it+1}\right)}_{\text{forecast revision}} + \epsilon^j_{it+1}$$

- $\beta \neq 0$ is inconsistent with rational expectations
  - Revisions are in forecasters' information set $\Rightarrow$ should not predict errors
- $\beta > 0 \Rightarrow$ revisions do not update "enough" $\Rightarrow$ **underreaction** Bouchaud et al. 19
- $\beta < 0 \Rightarrow$ revisions update "too much" $\Rightarrow$ **overreaction** Bordalo et al. 19
- Now a standard way of characterizing deviations from RE across datasets

Outline

Fact 1: Non-Linearity in Revisions (Sales Growth)



- Forecasts underreact **and** overreact within **same** variable and horizon

Fact 1: Non-Linearity in Revisions (22 Variables)

$$ERR^{j}_{it+1} = \alpha + \beta REV^{j}_{it} + \gamma REV^{j}_{it} \times (P^{+}_{ijt} + P^{-}_{ijt}) + \delta^{+} P^{+}_{ijt} + \delta^{-} P^{-}_{ijt} + e^{j}_{it+1}$$

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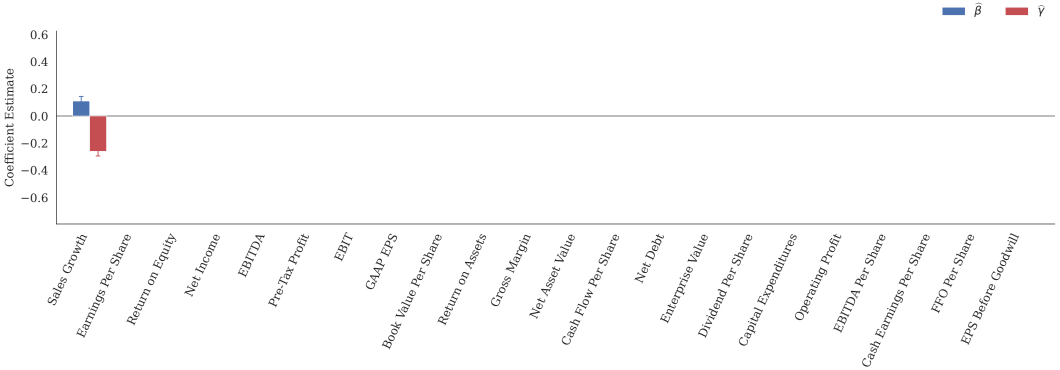

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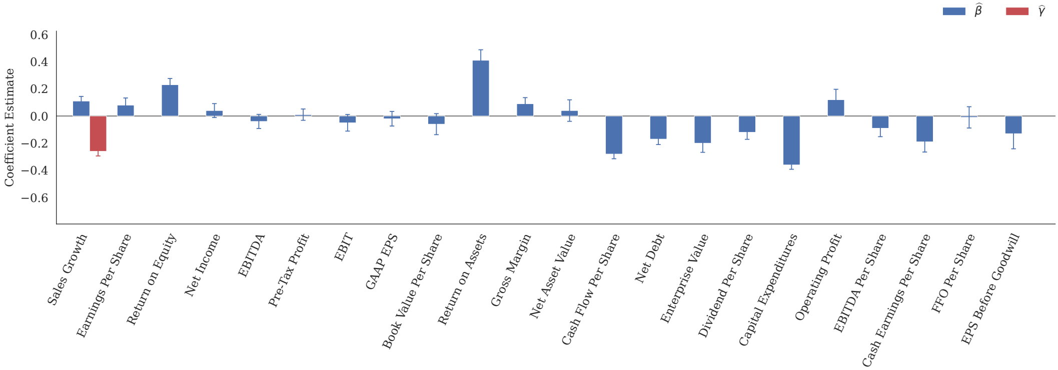

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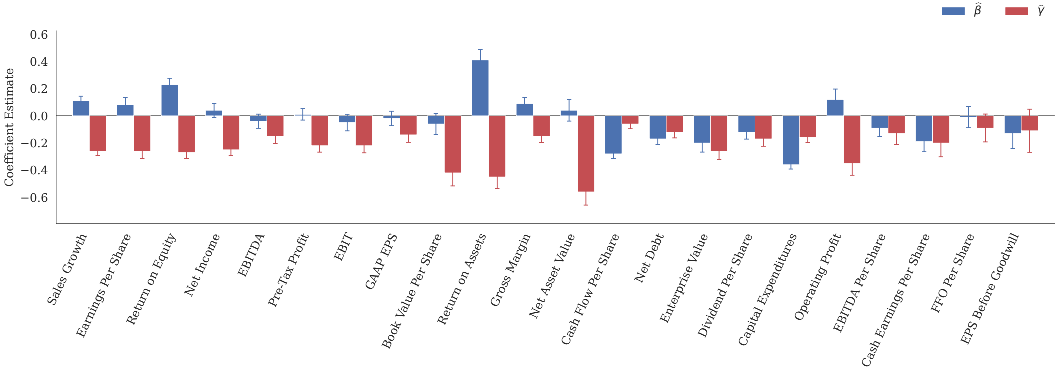

- Stronger overreaction/less underreaction in tails for **20 of 22** variables

▸ Robustness

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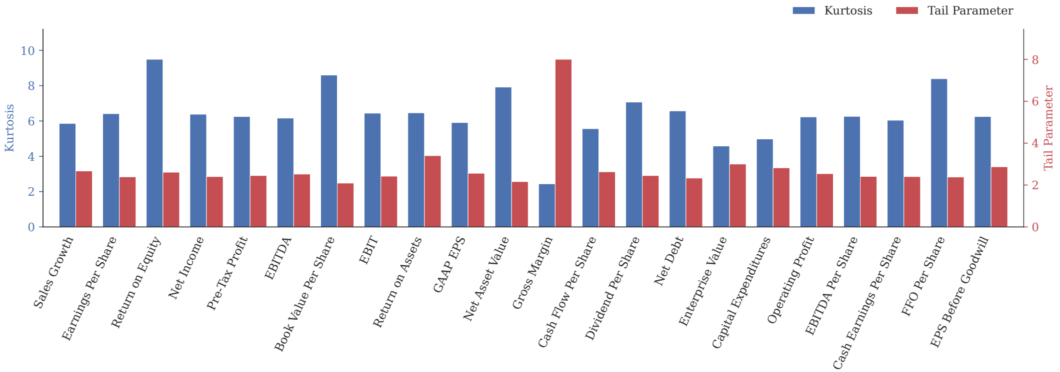
Fact 2: Realizations Have Fat Tails (22 Variables)

$$\log \mathrm{rank}(|g_{it}|) = \alpha_\nu - \nu \log |g_{it}| + e_{it}$$



- R^2 in power law regressions $>$ **99%** for all variables

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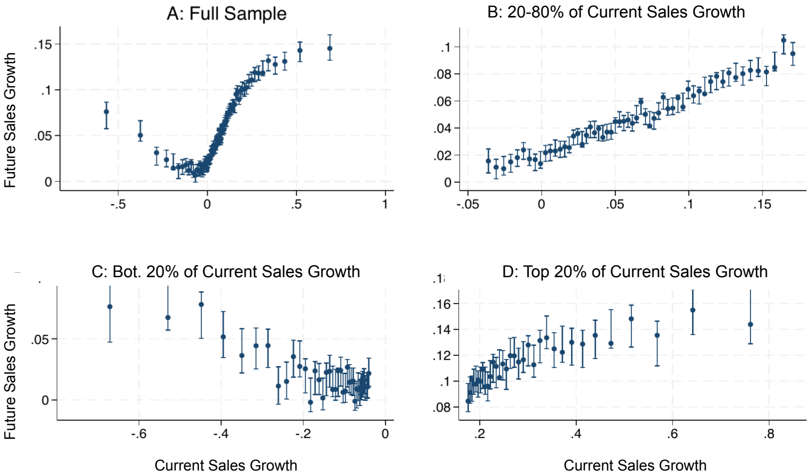
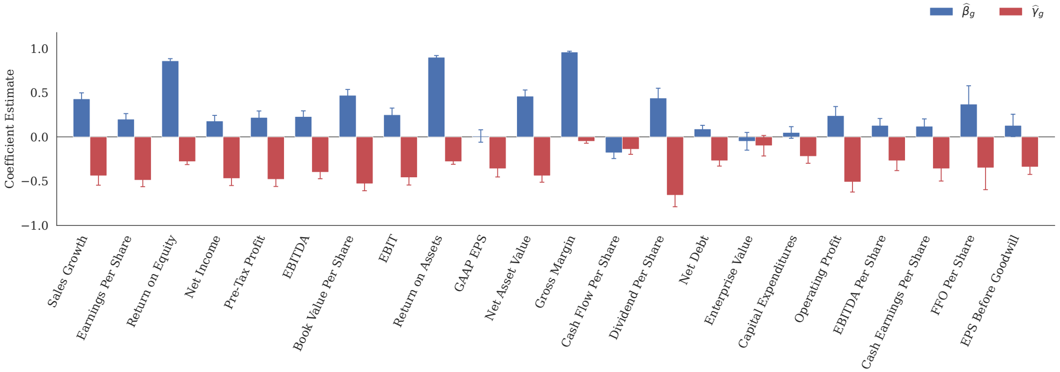
Fact 3: Expected Realizations Are Non-Linear (Sales Growth)



- Sales growth is persistent in the bulk, but attenuates or mean-reverts in the tails

Fact 3: Expected Realizations Are Non-Linear (22 Variables)

$$g_{it+1} = \alpha_g + \beta_g g_{it} + \gamma_g g_{it} \times (M_{it}^+ + M_{it}^-) + \delta_g^+ M_{it}^+ + \delta_g^- M_{it}^- + e_{it+1}$$



- Tail slope change $\widehat{\gamma}_g$ is negative for **all 22** variables

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- DGP for generic forecasting variable (dropping *i* subscripts):

$$\begin{aligned} g^*_{t+1} &= \rho g^*_t + \sigma_u u_{t+1} & P(u_t > x) &= o(x^{-\nu}) \\ g_{t+1} &= g^*_{t+1} + \sigma_\epsilon \epsilon_{t+1} & P(\epsilon_t > x) &\sim \Theta x^{-\nu} \quad \nu > 2 \end{aligned}$$

- g_t is a combination of persistent & transitory processes Bansal-Yaron 04, Lettau-Wachter 07
  - g^*_t = **unobservable** persistent latent state
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  - g^*_t = **unobservable** persistent latent state
  - ϵ_t = transitory shock with **power-law tail**
- Remarks:
  - If both shocks were Gaussian, rational expectation would be the Kalman filter
  - **Key**: u_t has thinner tails than ϵ_t, otherwise inconsistent with Fact 3

DGP Replicates Facts 2 and 3

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  - **Intuition**: moderate values of g_t likely reflect $g^*_t \Rightarrow$ likely persistent

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One Departure from Rational Expectations Replicates Fact 1

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- **Result**: Additionally, there exists a $R > 0$ where:

$$E\left(\text{error}_{t+1} \times \text{revision}_t \middle| |\text{revision}_t| < R\right) > 0$$

  - Overreaction in tails + unbiased on average $\Rightarrow$ underreaction in bulk $\Rightarrow$ Fact 1

Connection to Insensitivity to Signal Strength

- Evidence of insensitivity to **signal strength** in lab inference problems
  - Overreaction to weak $+$ underreaction to strong signals Augenblick et al. 24, Ba et al. 24
  - Challenge: how to generalize this behavior to a time-series setting?

Connection to Insensitivity to Signal Strength

- Evidence of insensitivity to **signal strength** in lab inference problems
  - Overreaction to weak + underreaction to strong signals Augenblick et al. 24, Ba et al. 24
  - Challenge: how to generalize this behavior to a time-series setting?

- **Result**: If u_t is normal,

$$\mathrm{var}(g_{t+1}|g_t) = \sigma_g^2 + \rho^2 \sigma_{g^*}^4 \frac{d^2}{dg^2} \log h(g_t)$$

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  - Tails: log density is convex $\Rightarrow$ weak signal, and have overreaction

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  - Bulk: log density is concave $\Rightarrow$ strong signal, and have underreaction
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- Ignoring fat tails looks like ignoring signal strength in time-series setting!

Quantitative Fit: Forecasts Partially Ignore Fat Tails

- Estimate sales growth DGP to match Facts #2 and #3 ▸ Parameters
- Shrink forecasts away from the Kalman default toward RE:

$$F_t^{\lambda} g_{t+h} = \lambda F_t g_{t+h} + (1 - \lambda) E_t g_{t+h}$$

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- Shrink forecasts away from the Kalman default toward RE:

$$F_t^{\lambda} g_{t+h} = \lambda F_t g_{t+h} + (1-\lambda) E_t g_{t+h} \qquad \widehat{\lambda} \approx 0.2$$

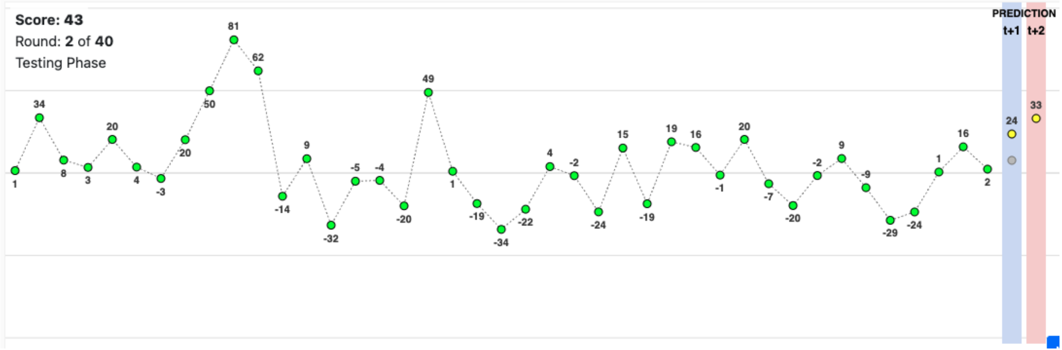
Moment	Data	Model	Model ($\lambda = 1$)
$\widehat{\beta}_g$	0.429	0.425	.
$\widehat{\gamma}_g$	-0.438	-0.437	.
Tail Parameter	2.670	2.670	.
P90-P10 g_t	2.146	2.147	.
P90-P10 $g_{t+1} - g_t$	2.595	2.594	.
P90-P10 $g_{t+3} - g_t$	2.844	2.842	.
$\widehat{\beta}$	0.106	0.144	1.816
$\widehat{\gamma}$	-0.265	-0.255	-2.861

Outline

1. Three Key Facts
   - Fact 1: Non-Linear Error-Revision Relationship
   - Fact 2: Fat Tails in the Distribution of Realizations
   - Fact 3: Expected Realizations are Non-Linear
2. Model of Expectations Formation
3. **Additional Evidence**
   - Forecasting Experiment
   - Macro Forecasts
   - Return Momentum
4. Conclusion

Outline

- Design follows Afrouzi et al. 23: participants make one and two-period forecasts
- 403 MTurk participants make 40 forecasts $\Rightarrow$ 16,120 observations
- DGPs: 1. rescaled estimated DGP $+$ 2. Gaussian AR1 with $\rho = 0.2$ Afrouzi et al. 23



Fat Tails Create Non-Linear Error-Revision Relationship

	Dependent Variable: Error_{it+1}	
	Afrouzi et al.	
	(1)	(2)
Revision_{it}	-0.42	-0.40
	(-21.56)	(-10.24)
Revision_{it} × Top/Bottom 20% of Revision $_{it}$		-0.06
		(-1.17)
Top 20% of Revision_{it}		-2.37
		(-0.98)
Bottom 20% of Revision_{it}		-7.57
		(-2.94)
Constant	-6.98	-5.22
	(-5.33)	(-4.03)
# of Obs.	6859	6859

- Non-linear relationship between errors and revisions arises with non-Gaussian DGP

Fat Tails Create Non-Linear Error-Revision Relationship

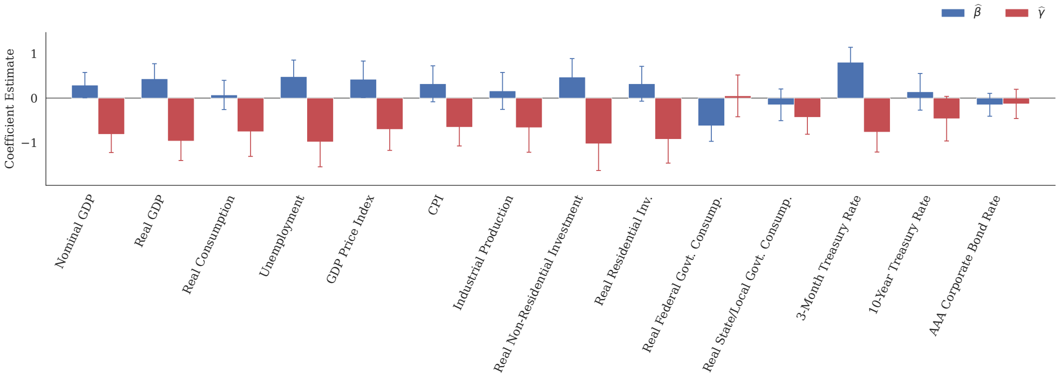
	Dependent Variable: Error_{it+1}			
	Afrouzi et al.		New Experiment	
	(1)	(2)	(3)	(4)
Revision_{it}	-0.42	-0.40	-0.35	-0.27
	(-21.56)	(-10.24)	(-20.51)	(-6.73)
$\text{Revision}_{it} \times$ Top/Bottom 20% of Revision_{it}		-0.06		-0.17
		(-1.17)		(-3.28)
Top 20% of Revision_{it}		-2.37		7.64
		(-0.98)		(3.55)
Bottom 20% of Revision_{it}		-7.57		-3.96
		(-2.94)		(-1.53)
Constant	-6.98	-5.22	-8.40	-9.14
	(-5.33)	(-4.03)	(-7.86)	(-7.40)
# of Obs.	6859	6859	15003	15003

- Non-linear relationship between errors and revisions arises with non-Gaussian DGP

Outline

Similar Non-Linearity in SPF Forecasts

$$ERR^j_{it+1} = \alpha + \beta REV^j_{it} + \gamma REV^j_{it} \times (P^+_{ijt} + P^-_{ijt}) + \delta^+ P^+_{ijt} + \delta^- P^-_{ijt} + e^j_{it+1}$$



- Non-linearity arises for all but one variable in (updated) Bordalo et al. 20 sample

Outline

- Campbell 91 + constant subjective discount rate links returns to revisions

$$\log(1+R_t) \approx \log(1+R_f+\pi) + (g_t - F_{t-1}g_t) + \sum_{k=1}^{\infty} c^k REV_t g_{t+k}$$

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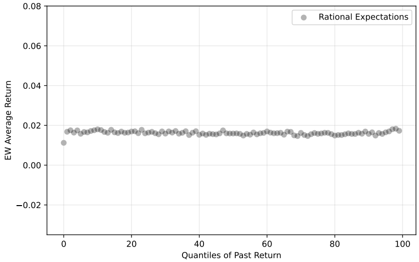
Model



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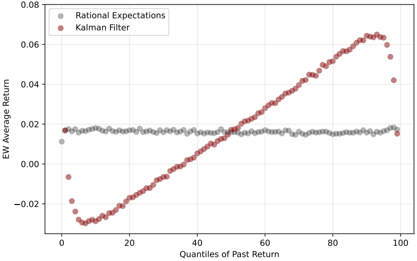

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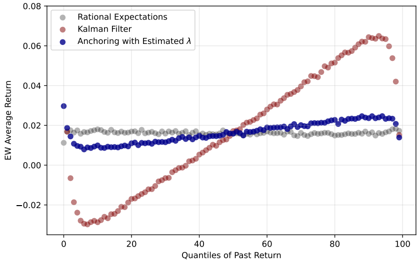

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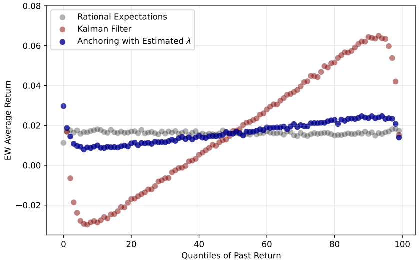
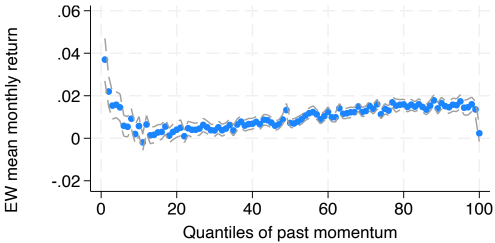


Data (Bottom Half Mkt Cap)



Outline

Conclusion

- Main fact: forecast errors are **non-linear** in forecast revisions across 22 variables
  - Less overreaction in the bulk, stronger overreaction in the tails
- One deviation from RE can explain this: **ignoring fat tails**
  - **Intuition**: extreme realizations are less persistent than forecasters realize
  - Provides a parsimonious model of under **and** overreaction **within a DGP**
  - Also consistent with experiments, macro forecasts, and asset prices
- Broader takeaways:
  1. **Non-Gaussian models of DGP** are helpful for understanding belief formation
  2. **Combining** experiments + surveys useful for assessing important features

THANK YOU!

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Additional Results and Robustness

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2. Does not reflect omitted Jensen's term: holds with **percent growth**

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4. Not a mixture of thin-tailed processes: robust to **firm-level volatility**
5. Driven by cross-section, not aggregate **mean** or **volatility**
6. Limited evidence of learning: does not vary with **analyst experience**

Back

Fat Tails: Robustness

1. Not a mixture of thin-tailed processes: robust to **firm-level volatility**
2. Driven by cross-section, not time-varying aggregate **volatility**
3. Does not reflect omitted Jensen's term: holds with **percent growth**

Back

Conditional Expectation: Robustness

1. Not a mixture of thin-tailed processes: robust to **firm-level volatility**
2. Driven by cross-section, not time-varying aggregate **volatility**
3. Does not reflect omitted Jensen's term: holds with **percent growth**
4. Gaussian approximation captures the **bulk**

Back

Quantitative Fit: Sales Growth DGP

- Four DGP parameters target Facts #2 and #3 with SMD
- λ targets the two error-revision moments in Fact #1

	ρ	σ_u	σ_ϵ	ν	λ
Estimate	0.784	0.336	1.058	2.411	0.197
Std. Error	0.020	0.018	0.046	0.037	0.024

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Fact 1: Consensus Forecasts

$Error_{it+1}$	Revision		Tail Slope		Top		Bottom		
Variable	$\widehat{\beta}$	t	$\widehat{\gamma}$	t	$\widehat{\delta+}$	t	$\widehat{\delta-}$	t	Obs.
Sales growth	0.23	5.34	-0.40	-8.31	0.01	0.92	-0.04	-11.76	145225
Return On Equity	0.40	7.63	-0.44	-6.86	0.00	2.01	-0.03	-10.09	128761
Earnings Per Share	0.02	0.49	-0.22	-4.82	-0.03	-2.58	-0.07	-6.14	126893
Net Income	-0.08	-1.79	-0.12	-2.77	-0.03	-1.76	-0.05	-4.69	112990
Pre-tax Profit	-0.07	-1.38	-0.13	-2.69	-0.02	-1.63	-0.04	-3.18	106455
Book Value Per Share	-0.21	-7.02	-0.28	-8.62	-0.01	-3.15	-0.04	-10.06	99986
EBITDA	-0.03	-1.29	-0.18	-5.02	-0.01	-0.92	-0.05	-3.94	98819
Net Asset Value	-0.31	-9.97	-0.23	-6.44	0.01	2.89	-0.01	-1.11	89954
Dividend Per Share	-0.00	-0.02	-0.30	-4.20	-0.00	-0.22	-0.04	-4.75	89927
Return On Assets	0.46	11.50	-0.51	-11.36	-0.01	-9.72	-0.02	-14.44	89807
EBIT	-0.16	-4.82	-0.08	-2.13	-0.02	-1.83	-0.03	-2.57	83770
GAAP EPS	-0.19	-4.19	0.01	0.23	-0.02	-1.12	-0.00	-0.30	82578
Capital Expenditures	-0.40	-25.03	-0.23	-9.78	0.10	6.04	-0.09	-5.38	78871
Gross Margin	0.08	2.39	-0.27	-6.58	0.00	3.16	-0.01	-7.67	77518
Cash Flow Per Share	-0.33	-23.51	0.03	1.50	0.01	1.05	0.02	2.94	70787
Net Debt	-0.23	-7.75	-0.00	-0.13	0.11	10.77	0.14	13.20	44693
Enterprise Value	-0.10	-1.50	-0.36	-5.79	0.09	7.82	-0.01	-1.13	43855
Operating Profit	-0.01	-0.15	-0.32	-4.52	-0.00	-0.36	-0.07	-3.93	43025
Cash Earnings Per Share	-0.39	-12.04	0.01	0.12	-0.02	-1.99	-0.02	-0.90	22777
EBITDA Per Share	-0.09	-1.91	-0.16	-2.56	-0.00	-0.26	-0.03	-2.78	22528
Funds From Operations Per Share	-0.00	-0.04	-0.23	-2.29	-0.00	-0.13	-0.04	-4.77	3548
EPS Before Goodwill	-0.29	-1.90	0.08	0.48	0.00	0.30	-0.02	-0.71	2646

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FACT 1: ASYMMETRIC TAIL SLOPES

Error_{it+1}	Revision		Top Slope		Bottom Slope		Top		Bottom		
Variable	$\widehat{\beta}$	t	$\widehat{\gamma^+}$	t	$\widehat{\gamma^-}$	t	$\widehat{\delta^+}$	t	$\widehat{\delta^-}$	t	Obs.
Sales growth	0.11	6.31	-0.26	-11.33	-0.27	-12.74	0.02	12.25	-0.02	-5.72	399809
Earnings Per Share	0.08	2.97	-0.24	-6.65	-0.27	-9.64	0.01	1.48	-0.08	-7.58	360011
Return On Equity	0.23	9.75	-0.37	-8.59	-0.22	-11.03	0.01	11.30	-0.01	-6.94	345082
Net Income	0.04	1.53	-0.21	-6.16	-0.27	-12.16	0.01	1.00	-0.07	-8.05	325230
EBITDA	-0.04	-1.51	-0.11	-2.29	-0.17	-6.55	0.00	1.43	-0.03	-3.58	280111
Pre-tax Profit	0.01	0.47	-0.16	-4.03	-0.27	-11.90	0.00	1.01	-0.07	-6.67	269883
EBIT	-0.05	-1.60	-0.14	-3.09	-0.26	-11.11	0.00	0.56	-0.05	-4.61	231455
GAAP EPS	-0.02	-0.73	-0.10	-2.84	-0.18	-7.87	-0.02	-1.01	-0.07	-3.97	171523
Book Value Per Share	-0.06	-1.52	-0.49	-8.97	-0.37	-7.56	0.02	5.10	-0.03	-5.08	170561
Return On Assets	0.41	10.35	-0.70	-11.25	-0.35	-9.63	-0.00	-0.77	-0.01	-6.80	157063
Gross Margin	0.09	3.90	-0.23	-6.44	-0.12	-4.28	0.00	5.18	-0.00	-3.61	150220
Net Asset Value	0.04	0.99	-0.41	-6.85	-0.68	-13.71	0.01	3.11	-0.05	-9.43	119885
Cash Flow Per Share	-0.28	-15.86	0.04	1.47	-0.15	-6.36	-0.01	-0.94	-0.07	-5.97	112323
Net Debt	-0.17	-8.31	0.13	5.69	-0.54	-21.33	-0.06	-5.67	-0.05	-4.02	90126
Enterprise Value	-0.20	-5.82	-0.01	-0.12	-0.44	-11.09	0.02	1.80	-0.07	-7.76	88791
Dividend Per Share	-0.12	-4.54	-0.18	-4.51	-0.16	-5.66	-0.01	-0.61	-0.03	-2.89	63840
Capital Expenditures	-0.36	-21.93	-0.14	-5.98	-0.19	-6.85	0.05	3.53	-0.07	-5.62	62099
Operating Profit	0.12	3.07	-0.31	-5.97	-0.37	-7.95	0.02	3.59	-0.07	-6.42	57756
EBITDA Per Share	-0.09	-2.84	-0.15	-2.37	-0.12	-2.72	0.03	3.45	-0.02	-1.67	16824
Cash Earnings Per Share	-0.19	-4.95	-0.15	-2.12	-0.24	-4.17	0.01	0.79	-0.04	-2.11	11578
Funds From Operations Per Share	-0.01	-0.25	-0.03	-0.46	-0.11	-1.90	0.00	1.01	-0.02	-2.09	9279
EPS Before Goodwill	-0.13	-2.29	-0.18	-1.46	-0.06	-0.71	0.06	2.61	-0.06	-3.33	2558

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Fact 1: Percent Growth

Error_{it+1}	Revision		Tail Slope		Top		Bottom		
Variable	$\widehat{\beta}$	t	$\widehat{\gamma}$	t	$\widehat{\delta+}$	t	$\widehat{\delta-}$	t	Obs.
Sales growth	0.10	5.67	-0.27	-13.42	0.03	7.86	-0.02	-3.65	399832
Earnings Per Share	0.00	0.05	-0.23	-7.17	0.06	4.11	-0.05	-2.06	364593
Net Income	-0.04	-1.67	-0.22	-7.25	0.06	4.14	-0.04	-1.89	329340
EBITDA	-0.08	-3.25	-0.13	-3.94	0.02	2.65	-0.01	-0.73	281746
Pre-tax Profit	-0.07	-3.18	-0.18	-6.45	0.05	3.26	-0.01	-0.52	273285
EBIT	-0.12	-4.30	-0.14	-3.88	0.04	3.11	0.00	0.24	234200
GAAP EPS	-0.11	-5.92	-0.15	-4.91	0.07	2.89	-0.02	-0.74	173552
Book Value Per Share	-0.11	-2.77	-0.40	-8.73	0.02	3.55	-0.04	-7.12	170925
Net Asset Value	-0.02	-0.60	-0.45	-8.97	0.04	6.39	-0.02	-2.17	120417
Cash Flow Per Share	-0.30	-17.77	-0.02	-1.12	0.00	0.37	0.00	0.07	113717
Net Debt	-0.09	-4.25	-0.04	-1.78	0.09	6.20	0.25	16.00	96338
Enterprise Value	-0.17	-4.60	-0.04	-0.89	0.05	5.12	0.06	4.64	88700
Dividend Per Share	-0.19	-6.65	-0.11	-3.04	-0.00	-0.27	-0.01	-0.61	64154
Capital Expenditures	-0.38	-25.04	-0.46	-24.24	0.29	15.58	-0.21	-15.90	62076
Operating Profit	0.02	0.79	-0.28	-6.18	0.06	3.32	-0.03	-1.76	58369
EBITDA Per Share	-0.15	-5.08	-0.12	-2.86	0.04	3.14	-0.02	-1.33	16905
Cash Earnings Per Share	-0.22	-7.03	-0.15	-2.73	0.01	0.95	-0.01	-0.27	11651
Funds From Operations Per Share	0.01	0.18	-0.14	-3.17	0.01	2.32	-0.02	-2.23	9324
EPS Before Goodwill	-0.18	-2.33	-0.18	-2.28	0.10	4.41	-0.11	-1.75	2560

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Fact 1: Firm-Level Volatility

Error_{it+1}	Revision		Tail Slope		Top		Bottom		
Variable	$\widehat{\beta}$	t	$\widehat{\gamma}$	t	$\widehat{\delta^+}$	t	$\widehat{\delta^-}$	t	Obs.
Sales growth	0.11	6.81	-0.33	-16.89	0.20	9.95	-0.16	-9.31	303623
Earnings Per Share	0.03	1.49	-0.26	-7.67	0.15	7.12	-0.12	-3.85	272283
Return On Equity	0.18	11.53	-0.31	-15.35	0.13	6.05	-0.17	-8.09	248053
Net Income	-0.02	-1.11	-0.24	-7.07	0.17	7.12	-0.10	-2.90	242091
EBITDA	-0.03	-1.67	-0.29	-11.04	0.19	7.35	-0.13	-4.85	205224
Pre-tax Profit	-0.00	-0.18	-0.32	-11.37	0.20	8.94	-0.13	-3.43	196802
EBIT	-0.06	-2.21	-0.41	-10.22	0.30	12.92	-0.21	-4.35	159223
Book Value Per Share	-0.12	-2.92	-0.40	-8.27	0.16	5.64	-0.17	-5.36	119693
GAAP EPS	-0.03	-1.06	-0.20	-5.21	0.15	4.30	-0.06	-1.97	110490
Return On Assets	0.23	8.95	-0.39	-11.66	0.00	0.07	-0.38	-9.18	95610
Gross Margin	0.11	5.23	-0.25	-11.33	0.12	5.29	-0.17	-9.62	85574
Cash Flow Per Share	-0.27	-16.27	-0.21	-9.07	0.27	11.26	-0.11	-2.68	74669
Net Asset Value	-0.07	-2.03	-0.62	-10.91	0.41	16.04	-0.24	-4.08	70815
Net Debt	-0.15	-5.53	-0.22	-5.64	0.38	8.35	0.25	6.64	58422
Enterprise Value	-0.15	-3.65	-0.46	-11.48	0.55	16.25	-0.11	-2.99	43748
Dividend Per Share	-0.21	-8.59	-0.15	-4.54	0.12	4.02	-0.07	-2.22	38038
Capital Expenditures	-0.33	-16.59	-0.13	-5.00	0.10	1.66	-0.08	-2.57	33460
Operating Profit	0.15	2.85	-0.46	-7.14	0.26	6.52	-0.12	-2.47	22761
Funds From Operations Per Share	0.01	0.18	-0.11	-1.86	0.11	3.03	-0.11	-2.27	6941
Cash Earnings Per Share	-0.25	-5.81	-0.26	-4.16	0.23	3.90	-0.28	-3.31	4483
EBITDA Per Share	-0.04	-1.08	-0.36	-6.07	0.25	4.43	-0.24	-5.92	4068

Back

FACT 1: REMOVING AGGREGATE MEAN

Error_{it+1}	Revision		Tail Slope		Top		Bottom		
Variable	$\widehat{\beta}$	t	$\widehat{\gamma}$	t	$\widehat{\delta^+}$	t	$\widehat{\delta^-}$	t	Obs.
Sales growth	0.08	3.35	-0.25	-11.65	0.02	6.84	-0.02	-6.34	399799
Earnings Per Share	0.07	2.31	-0.24	-8.60	0.01	0.84	-0.06	-6.39	359967
Return On Equity	-0.00	-0.00	-0.24	-0.66	0.10	1.04	-1.31	-1.42	340601
Net Income	0.02	0.87	-0.22	-10.67	0.01	0.90	-0.05	-5.34	325194
EBITDA	-0.03	-1.23	-0.15	-6.43	0.01	1.25	-0.02	-3.11	280094
Pre-tax Profit	-0.01	-0.30	-0.19	-7.03	0.02	1.87	-0.04	-3.76	269840
EBIT	-0.04	-1.39	-0.23	-9.58	0.03	3.25	-0.02	-2.40	231446
GAAP EPS	-0.05	-0.98	-0.10	-2.08	-0.01	-0.94	-0.04	-2.81	171544
Book Value Per Share	-0.10	-2.11	-0.39	-7.23	0.02	5.64	-0.03	-6.55	170543
Return On Assets	0.21	3.08	-0.24	-3.20	-0.00	-1.61	-0.01	-4.42	157075
Gross Margin	0.09	4.65	-0.14	-6.09	0.00	2.75	-0.00	-6.62	150215
Net Asset Value	-0.03	-0.65	-0.49	-8.02	0.04	11.91	-0.02	-3.21	119887
Cash Flow Per Share	-0.25	-16.94	-0.09	-5.15	0.05	4.70	-0.00	-0.30	112347
Net Debt	-0.15	-6.46	-0.14	-6.44	0.11	10.47	0.11	11.70	90108
Enterprise Value	-0.20	-5.78	-0.26	-7.32	0.09	10.94	-0.02	-2.10	88772
Dividend Per Share	-0.16	-4.84	-0.15	-4.92	-0.00	-0.59	-0.03	-3.87	63847
Capital Expenditures	-0.37	-21.85	-0.16	-7.53	0.04	4.03	-0.06	-6.36	62094
Operating Profit	0.04	0.98	-0.26	-6.43	0.04	3.97	-0.04	-5.15	57758
EBITDA Per Share	-0.04	-0.91	-0.16	-3.31	0.01	0.80	-0.02	-1.36	16820
Cash Earnings Per Share	-0.21	-5.63	-0.18	-3.36	0.04	2.56	-0.02	-1.01	11578
Funds From Operations Per Share	-0.01	-0.15	-0.10	-1.74	0.01	1.45	-0.01	-1.32	9280
EPS Before Goodwill	-0.13	-2.12	-0.07	-0.92	0.01	0.65	-0.06	-2.40	2558

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Error_{it+1}	Revision		Tail Slope		Top		Bottom		
Variable	$\widehat{\beta}$	t	$\widehat{\gamma}$	t	$\widehat{\delta^+}$	t	$\widehat{\delta^-}$	t	Obs.
Sales growth	0.11	5.46	-0.27	-12.06	0.06	4.90	-0.09	-7.87	399555
Earnings Per Share	0.08	2.53	-0.25	-10.00	0.02	1.26	-0.11	-5.65	359817
Return On Equity	0.21	8.94	-0.22	-8.29	0.00	1.05	-0.03	-5.32	344222
Net Income	0.03	1.42	-0.24	-13.95	0.02	1.25	-0.10	-5.90	325134
EBITDA	-0.04	-1.34	-0.14	-5.99	0.02	1.18	-0.05	-2.64	279943
Pre-tax Profit	0.00	0.09	-0.21	-10.90	0.03	1.79	-0.08	-3.97	269692
EBIT	-0.04	-1.37	-0.22	-9.49	0.04	2.50	-0.06	-2.93	231398
GAAP EPS	-0.02	-0.91	-0.13	-4.67	-0.00	-0.03	-0.06	-2.21	171482
Book Value Per Share	-0.07	-1.71	-0.41	-7.92	0.03	2.49	-0.12	-6.63	170367
Return On Assets	0.50	8.27	-0.52	-8.59	-0.08	-4.80	-0.14	-8.02	156935
Gross Margin	0.10	4.23	-0.15	-5.62	0.00	1.33	-0.02	-5.75	150124
Net Asset Value	0.02	0.62	-0.55	-11.89	0.11	8.67	-0.11	-5.41	119865
Cash Flow Per Share	-0.28	-15.30	-0.08	-4.21	0.08	4.39	-0.01	-0.52	112303
Net Debt	-0.17	-8.09	-0.12	-5.44	0.20	10.25	0.20	8.31	90095
Enterprise Value	-0.19	-5.20	-0.28	-8.53	0.26	12.27	-0.03	-1.03	88775
Dividend Per Share	-0.14	-5.46	-0.19	-7.40	0.00	0.01	-0.08	-4.87	63788
Capital Expenditures	-0.37	-23.67	-0.16	-6.75	0.12	4.93	-0.10	-4.19	62079
Operating Profit	0.12	3.29	-0.35	-8.27	0.07	3.57	-0.11	-4.80	57718
EBITDA Per Share	-0.09	-3.02	-0.13	-3.47	0.04	1.88	-0.06	-1.98	16816
Cash Earnings Per Share	-0.18	-4.54	-0.19	-3.41	0.04	1.20	-0.04	-0.90	11563
Funds From Operations Per Share	0.05	1.62	-0.15	-2.72	0.03	1.19	-0.06	-1.67	9231
EPS Before Goodwill	-0.11	-2.23	-0.10	-1.19	0.06	1.26	-0.13	-2.30	2552

Back

Fact 1: Variation with Analyst Experience

Error_{it+1}	Exp. Q1		Exp. Q2		Exp. Q3		Exp. Q4	
Variable	$\widehat{\gamma}$	t	$\widehat{\gamma}$	t	$\widehat{\gamma}$	t	$\widehat{\gamma}$	t
Sales growth	-0.28	-16.28	-0.28	-14.70	-0.25	-11.09	-0.25	-7.54
Earnings Per Share	-0.27	-8.91	-0.25	-6.12	-0.27	-7.83	-0.26	-6.78
Return On Equity	-0.27	-9.27	-0.30	-12.03	-0.27	-9.72	-0.22	-4.86
Net Income	-0.24	-8.63	-0.24	-6.00	-0.26	-10.26	-0.24	-8.32
EBITDA	-0.16	-5.06	-0.15	-4.81	-0.13	-3.53	-0.13	-4.18
Pre-tax Profit	-0.26	-7.28	-0.20	-5.70	-0.25	-8.57	-0.19	-6.45
EBIT	-0.23	-7.12	-0.19	-6.82	-0.21	-4.97	-0.24	-6.00
GAAP EPS	-0.13	-3.66	-0.12	-3.58	-0.14	-4.52	-0.15	-3.32
Book Value Per Share	-0.40	-8.37	-0.40	-6.14	-0.42	-7.13	-0.46	-5.52
Return On Assets	-0.48	-9.55	-0.43	-6.74	-0.43	-7.70	-0.37	-7.01
Gross Margin	-0.16	-4.52	-0.15	-2.76	-0.14	-2.91	-0.17	-3.91
Net Asset Value	-0.59	-14.20	-0.57	-8.99	-0.49	-8.11	-0.59	-6.65
Cash Flow Per Share	-0.03	-0.71	-0.05	-1.96	-0.08	-3.10	-0.10	-2.18
Net Debt	-0.13	-3.98	-0.08	-2.41	-0.11	-2.15	-0.18	-5.92
Enterprise Value	-0.23	-4.56	-0.26	-5.86	-0.30	-8.44	-0.26	-5.61
Dividend Per Share	-0.22	-4.31	-0.15	-3.49	-0.13	-2.44	-0.16	-2.90
Capital Expenditures	-0.15	-6.30	-0.13	-3.79	-0.10	-2.82	-0.26	-6.44
Operating Profit	-0.33	-7.17	-0.32	-4.20	-0.36	-5.86	-0.39	-5.49
EBITDA Per Share	-0.22	-2.89	-0.02	-0.29	-0.14	-1.97	-0.11	-1.70
Cash Earnings Per Share	-0.23	-3.43	-0.12	-0.86	-0.24	-1.94	-0.15	-3.34
Funds From Operations Per Share	0.03	0.33	-0.14	-1.80	-0.09	-1.08	-0.16	-1.34
EPS Before Goodwill	-0.02	-0.22	-0.03	-0.17	-0.06	-0.22	-0.30	-2.31

Back

Fat Tails: Adjusting for Firm-Level Volatility

Variable	Transformation	Kurtosis	Tail Parameter	R^2	Obs.
Sales growth	Growth	3.40	4.14	0.99	87500
Earnings Per Share	Growth	3.70	4.17	0.98	80346
Return On Equity	Level	3.76	4.15	1.00	73100
Net Income	Growth	3.60	4.25	0.98	69126
Pre-tax Profit	Growth	3.66	4.13	0.98	59902
EBITDA	Growth	3.40	4.40	0.98	59770
Book Value Per Share	Growth	3.64	3.94	0.99	50930
EBIT	Growth	3.49	4.46	0.98	44622
Return On Assets	Level	4.01	3.92	0.99	38226
GAAP EPS	Growth	3.47	4.59	0.98	36745
Cash Flow Per Share	Growth	3.23	4.58	0.98	33233
Net Asset Value	Growth	3.32	3.86	0.99	33099
Gross Margin	Level	5.38	3.17	0.99	27309
Net Debt	Growth	3.80	3.76	0.98	24395
Dividend Per Share	Growth	4.19	3.93	0.99	23469
Enterprise Value	Growth	2.80	5.07	0.98	18837
Capital Expenditures	Growth	2.80	5.26	0.98	17345
Operating Profit	Growth	3.60	4.24	0.98	10142
EBITDA Per Share	Growth	3.19	4.92	0.99	3089
Cash Earnings Per Share	Growth	3.10	5.08	0.97	2975
Funds From Operations Per Share	Growth	4.02	3.88	0.97	2250

Back

Fat Tails: Adjusting for Aggregate Volatility

Variable	Transformation	Kurtosis	Tail Parameter	R^2	Obs.
Sales growth	Growth	5.84	2.70	0.99	140481
Earnings Per Share	Growth	6.34	2.42	0.99	129735
Return On Equity	Level	7.62	2.76	1.00	127268
Net Income	Growth	6.28	2.42	0.99	113861
Pre-tax Profit	Growth	6.18	2.47	0.99	102386
EBITDA	Growth	6.07	2.53	0.99	99731
Book Value Per Share	Growth	8.38	2.13	1.00	89882
EBIT	Growth	6.36	2.48	0.99	82099
Return On Assets	Level	6.02	3.16	0.99	79477
GAAP EPS	Growth	5.88	2.57	0.99	74212
Net Asset Value	Growth	7.84	2.17	0.99	71494
Gross Margin	Level	2.47	7.21	0.98	64481
Cash Flow Per Share	Growth	5.52	2.64	0.99	62446
Dividend Per Share	Growth	6.87	2.50	0.99	49401
Net Debt	Growth	6.57	2.33	0.99	48338
Enterprise Value	Growth	4.44	3.12	0.99	48140
Capital Expenditures	Growth	4.96	2.82	0.99	44623
Operating Profit	Growth	6.08	2.54	0.99	33033
EBITDA Per Share	Growth	6.19	2.38	0.99	15344
Cash Earnings Per Share	Growth	6.01	2.44	0.99	10012
Funds From Operations Per Share	Growth	7.11	2.33	1.00	3399
EPS Before Goodwill	Growth	5.70	3.07	0.98	2413

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Fat Tails: Percent Growth

Variable	Transformation	Kurtosis	Tail Parameter	R^2	Obs.
Sales growth	Growth	8.27	2.35	1.00	140482
Earnings Per Share	Growth	25.66	1.29	1.00	129736
Net Income	Growth	27.02	1.29	1.00	113862
Pre-tax Profit	Growth	23.34	1.36	1.00	102386
EBITDA	Growth	14.73	1.89	1.00	99730
Book Value Per Share	Growth	9.80	2.26	0.99	89884
EBIT	Growth	19.63	1.52	1.00	82100
GAAP EPS	Growth	28.02	1.16	1.00	74221
Net Asset Value	Growth	11.92	2.08	1.00	71494
Cash Flow Per Share	Growth	23.00	1.31	1.00	62446
Dividend Per Share	Growth	15.61	1.66	0.99	49401
Net Debt	Growth	28.51	1.17	1.00	48339
Enterprise Value	Growth	9.01	2.06	1.00	48140
Capital Expenditures	Growth	16.21	1.60	1.00	44624
Operating Profit	Growth	19.50	1.55	1.00	33033
EBITDA Per Share	Growth	13.53	2.01	0.99	15344
Cash Earnings Per Share	Growth	11.62	2.05	0.99	10013
Funds From Operations Per Share	Growth	7.36	2.29	1.00	3399
EPS Before Goodwill	Growth	23.33	1.63	0.99	2414

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g_{it+1}	g_{it}		Tail Slope		Top		Bottom		
Variable	$\widehat{\beta}_g$	t	$\widehat{\gamma}_g$	t	$\widehat{\delta_g^+}$	t	$\widehat{\delta_g^-}$	t	Obs.
Sales growth	0.43	11.87	-0.36	-6.89	0.63	6.90	0.10	3.80	87500
Earnings Per Share	0.30	9.77	-0.53	-12.58	0.77	12.90	0.10	2.42	80346
Return On Equity	0.89	74.58	-0.03	-2.00	0.22	2.73	0.13	6.02	73100
Net Income	0.30	10.46	-0.51	-11.63	0.75	10.10	0.12	2.72	69126
Pre-tax Profit	0.30	9.28	-0.54	-13.47	0.79	12.29	0.09	2.39	59902
EBITDA	0.28	9.90	-0.40	-10.64	0.64	9.13	0.10	3.04	59770
Book Value Per Share	0.51	17.75	-0.45	-13.20	0.76	14.56	0.16	6.28	50930
EBIT	0.33	10.13	-0.53	-10.57	0.81	9.82	0.08	2.00	44622
Return On Assets	0.87	51.63	0.06	3.60	-0.20	-2.41	0.06	2.29	38226
GAAP EPS	0.12	3.74	-0.46	-11.98	0.72	11.63	0.11	2.33	36745
Cash Flow Per Share	-0.13	-3.89	-0.16	-4.44	0.36	9.43	0.14	5.13	33233
Net Asset Value	0.49	15.54	-0.40	-9.22	0.74	10.05	0.06	1.71	33099
Gross Margin	1.00	293.41	-0.01	-2.07	0.07	2.78	0.21	6.66	27309
Net Debt	0.12	4.18	-0.23	-6.82	0.34	9.23	0.09	2.23	24395
Dividend Per Share	0.44	10.73	-0.51	-9.41	0.85	9.68	0.15	4.85	23469
Enterprise Value	-0.03	-0.66	-0.16	-2.64	0.28	3.48	0.02	0.31	18837
Capital Expenditures	0.03	0.87	-0.11	-2.27	0.15	2.10	-0.07	-1.05	17345
Operating Profit	0.22	4.03	-0.48	-6.28	0.77	7.66	0.06	1.03	10142
EBITDA Per Share	0.30	4.89	-0.29	-2.81	0.55	2.64	0.22	1.88	3089
Cash Earnings Per Share	0.12	1.97	-0.29	-2.53	0.52	2.88	0.19	1.81	2975
Funds From Operations Per Share	0.39	5.53	-0.39	-3.50	0.51	3.84	-0.10	-1.01	2250

Back

CONDITIONAL EXPECTATION: ADJUSTING FOR AGGREGATE VOLATILITY

g_{it+1}	g_{it}		Tail Slope		Top		Bottom		
Variable	$\widehat{\beta}_g$	t	$\widehat{\gamma}_g$	t	$\widehat{\delta_g^+}$	t	$\widehat{\delta_g^-}$	t	Obs.
Sales growth	0.42	12.19	-0.41	-8.60	0.32	6.58	0.01	0.56	140481
Earnings Per Share	0.20	5.75	-0.47	-15.52	0.32	10.57	-0.03	-1.04	129735
Return On Equity	0.44	2.30	0.08	0.41	0.02	0.31	-0.03	-0.68	127268
Net Income	0.19	5.96	-0.45	-13.33	0.33	9.35	-0.02	-0.63	113861
Pre-tax Profit	0.19	5.25	-0.43	-10.98	0.34	10.26	-0.01	-0.28	102386
EBITDA	0.22	7.38	-0.39	-11.49	0.31	8.89	-0.01	-0.38	99731
Book Value Per Share	0.47	14.84	-0.51	-13.69	0.23	10.52	-0.08	-5.45	89882
EBIT	0.24	5.58	-0.44	-10.61	0.30	8.28	-0.01	-0.36	82099
Return On Assets	0.64	4.03	-0.04	-0.20	0.10	0.72	0.04	0.77	79477
GAAP EPS	0.02	0.34	-0.37	-8.35	0.28	10.25	0.00	0.01	74212
Net Asset Value	0.47	13.30	-0.45	-12.15	0.24	12.54	-0.08	-4.83	71494
Gross Margin	0.90	16.42	-0.23	-1.65	0.69	1.73	0.53	1.80	64481
Cash Flow Per Share	-0.18	-6.21	-0.13	-4.34	0.18	7.33	0.07	3.56	62446
Dividend Per Share	0.43	7.51	-0.65	-11.34	0.35	9.69	-0.06	-2.07	49401
Net Debt	0.08	4.18	-0.27	-8.77	0.24	10.35	0.00	0.18	48338
Enterprise Value	-0.06	-1.19	-0.09	-1.43	0.13	2.35	-0.03	-0.98	48140
Capital Expenditures	0.06	1.67	-0.22	-5.86	0.11	4.71	-0.10	-3.42	44623
Operating Profit	0.24	4.77	-0.49	-10.77	0.43	10.51	-0.02	-0.46	33033
EBITDA Per Share	0.12	2.67	-0.24	-3.11	0.13	2.12	0.01	0.09	15344
Cash Earnings Per Share	0.08	2.12	-0.34	-5.68	0.21	4.36	-0.03	-0.86	10012
Funds From Operations Per Share	0.46	4.25	-0.46	-3.72	0.22	2.09	-0.09	-1.08	3399
EPS Before Goodwill	0.21	2.09	-0.54	-3.97	0.40	3.45	-0.08	-1.00	2413

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Conditional Expectation: Percent Growth

g_{it+1}	g_{it}		Tail Slope		Top		Bottom		
Variable	$\widehat{\beta_g}$	t	$\widehat{\gamma_g}$	t	$\widehat{\delta_g^+}$	t	$\widehat{\delta_g^-}$	t	Obs.
Sales growth	0.44	12.74	-0.39	-8.69	0.11	7.31	0.02	2.17	140482
Earnings Per Share	0.14	4.50	-0.17	-5.25	0.14	7.03	0.35	8.50	129736
Net Income	0.14	5.26	-0.17	-6.14	0.15	7.09	0.37	8.16	113862
Pre-tax Profit	0.19	6.40	-0.20	-7.08	0.15	7.04	0.33	7.33	102386
EBITDA	0.21	6.68	-0.24	-6.75	0.12	7.32	0.13	5.53	99730
Book Value Per Share	0.45	14.10	-0.52	-15.55	0.10	14.80	-0.01	-2.31	89884
EBIT	0.23	6.87	-0.25	-6.98	0.13	5.89	0.23	5.69	82100
GAAP EPS	-0.05	-1.54	0.01	0.40	0.08	3.16	0.60	9.23	74221
Net Asset Value	0.46	14.08	-0.45	-13.88	0.10	18.03	-0.01	-1.19	71494
Cash Flow Per Share	-0.20	-6.26	0.16	5.62	0.01	0.79	0.51	15.81	62446
Dividend Per Share	0.36	7.92	-0.45	-8.44	0.17	9.34	0.13	4.57	49401
Net Debt	0.07	2.47	-0.10	-2.97	0.11	6.62	0.30	11.81	48339
Enterprise Value	-0.06	-1.19	0.00	0.03	0.05	1.99	0.09	6.29	48140
Capital Expenditures	0.04	1.37	-0.10	-3.37	0.09	8.06	0.11	4.57	44624
Operating Profit	0.19	4.49	-0.22	-5.24	0.17	7.36	0.25	7.34	33033
EBITDA Per Share	0.11	3.09	-0.13	-2.72	0.07	2.51	0.13	2.90	15344
Cash Earnings Per Share	0.08	2.01	-0.20	-3.90	0.10	4.60	0.14	5.47	10013
Funds From Operations Per Share	0.34	3.53	-0.32	-2.54	0.05	2.45	-0.01	-0.39	3399
EPS Before Goodwill	0.13	2.37	-0.16	-2.37	0.09	1.43	0.17	3.53	2414

Back

Variable	R^2	Obs.
Sales growth	0.954	87500
Earnings Per Share	0.939	80345
Return On Equity	0.831	73100
Net Income	0.949	69126
Pre-tax Profit	0.940	59903
EBITDA	0.951	59771
Book Value Per Share	0.950	50929
EBIT	0.941	44620
Return On Assets	0.717	38226
GAAP EPS	0.908	36745
Cash Flow Per Share	0.929	33233
Net Asset Value	0.902	33099
Gross Margin	0.508	27309
Net Debt	0.894	24396
Dividend Per Share	0.893	23469
Enterprise Value	0.776	18838
Capital Expenditures	0.826	17344
Operating Profit	0.771	10142
EBITDA Per Share	0.631	3089
Cash Earnings Per Share	0.600	2976
Funds From Operations Per Share	0.583	2249

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